Worksheet 5-10 & 5-11 A Optimization Problems #1
AP Calculus Name_____

Solve each problem.

1.) A manufacturing company has determined that the total cost of producing an item can be determined from the equation $C = 8x^2 - 176x + 180$, where x is the number of units that the company makes. How many units should the company manufacture in order to minimize the cost (to the nearest whole number)?

2.) Determine the maximum area of a rectangular dog pen that can be enclosed with 200 meters of fence.

3.) Two numbers add up to 40. Find the numbers that maximize their product.

4.) A rectangle has a perimeter of 71 feet. What length and width should it have so that its area is a maximum? What is its maximum area?

5.) Find two positive numbers that minimize the sum of twice the first number plus the second if the product of the two numbers is 288.

6.) The number of flies around a dropped piece of bread is modeled by the formula $P(t) = \frac{1000t}{10+t^2}$ where t is measured in hours. How long will it take for the number of flies to be at a maximum? What is the maximum size of the population of flies (to the nearest fly)?